

UT Austin Numerical I Prelim Solutions

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This document contains solutions to the Numerical I prelim problems from 2015 to 2025. For those self-studying for this exam, the textbook used in the Numerical I course is *Numerical Mathematics* by Quarteroni, Sacco, and Saleri, and this textbook alone is sufficient to prepare for nearly all concepts on the prelim exam.

Disclaimer: I did not take the Numerical I course. These solutions are informed by my reading of the textbook and studying the previous prelim exams, so I may be wrong on some things here. If you think something on here is incorrect, please reach out!

Structure

In recent years, the Numerical I prelim has had a predictable structure. It consists of 3 problems, each with 3 parts, and will cover these topics (along with the corresponding chapters from the textbook):

Problem 1: Numerical Linear Algebra (Chapters 1, 3, 4, and 5)

Problem 2: Nonlinear Equations / Optimization (Chapters 6 and 7)

Problem 3: Numerical Integration / Polynomial Interpolation (Chapters 8, 9, and 10)

Many concepts in these chapters have not appeared on any previous preliminary exam. I am not sure which of these were skipped in the Numerical I course, so skip these untested concepts at your own risk.

Conventions

For all solutions, I will first defer to the notation of the problem as it's given in the prelim. Otherwise, I will defer to the textbook's notation. This will cause some inconsistency with things like iterates for Newton's method; the prelims often use subscripts to denote the iteration number, while the textbook puts this in a superscript.

Terminology

- The *subdiagonal* (resp. *superdiagonal*) of a square matrix $A \in \mathbb{R}^{n \times n}$ refers to the entries in the diagonal below (resp. above) the main diagonal of A . That is, it is the elements $a_{2,1}, a_{3,2}, \dots, a_{n,n-1}$ (resp. $a_{1,2}, a_{2,3}, \dots, a_{n-1,n}$).
- A matrix is in *upper Hessenberg form* if all entries below its subdiagonal are zero.

Eigenvalues of a Matrix

We will try to use the same notation that the book uses:

For A a square matrix:

- $\sigma(A) = \{\text{eigenvalues of } A\}$ is called the *spectrum* of A .
- $\rho(A) = \max_{\lambda \in \sigma(A)} |\lambda|$ is called the *spectral radius* of A .

Square Systems

When writing “consider a square system”, we assume the following setup:

- $A \in \mathbb{R}^{n \times n}$
- $b \in \mathbb{R}^n$
- x is a solution to $Ax = b$ (note: when describing iterative methods to find x , we do not assume a solution exists).
- For an iterative linear solver, I will use B with a subscript of the first letter of the solver’s name; for example, B_J will refer to the iteration matrix of the Jacobi method.
- For iterative linear solvers, $r^{(k)} := b - Ax^{(k)}$ will denote the residual at step k .
- We will let D denote the matrix containing the diagonal entries of A (and zero everywhere else), while E and F are respectively the lower and upper triangular parts of $D - A$ (which are, once again, zero everywhere else).

Overdetermined/Underdetermined Systems

Writing “consider an overdetermined system” will assume this setup:

- $m, n \in \mathbb{N}$ with $m \geq n$
- $A \in \mathbb{R}^{m \times n}$
- $b \in \mathbb{R}^m$

- $x^* \in \mathbb{R}^n$ is a minimum to $\|Ax - b\|_2^2$ and is a least-squares solution we want to find.

Likewise, any use of “consider an underdetermined system” will use the same notation as above, but with $m < n$.

Nonlinear Systems

Unless specified otherwise, all systems are assumed to be linear. When we specify by saying “consider a nonlinear system”, we assume the following setup:

- $n \in \mathbb{N}$
- $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ (not necessarily linear)
- The system is always written without loss of generality as $f(x) = 0$.
- x^* is a solution to the system, and any iterative methods that converge will converge to x^* .

Integration

All integration problems are over an interval in one dimension. If not specified, the endpoints of this interval are denoted by a and b , with $a < b$.

- $I(f)$ denotes the true integral $\int_a^b f(x) dx$, which is always assumed to exist.
- $I_n(f)$ denotes an approximation of the integral with $n + 1$ quadrature nodes. The exact type of approximation being used will depend on context, and any quadrature nodes being used are omitted from the notation but understood. For example, we may use $I_1(f)$ to denote a trapezoidal rule, or a Gaussian quadrature rule with two nodes.
- $I_{k,n}(f)$ denotes a composite approximation of $I(f)$ with n subintervals, and $k + 1$ points in each subinterval. For example, $I_{0,n}(f)$ may denote a composite midpoint rule with n subintervals.
- $E_n(f) := I(f) - I_n(f)$.
- $E_{k,n}(f) := I(f) - I_{k,n}(f)$.

January 2025

Problem 1

(a)

For a matrix $A \in \mathbb{R}^{n \times n}$, a vector $b \in \mathbb{R}^n$, and $m \in \mathbb{N}$, the Krylov subspace of A and b of order m is

$$K_m(A; b) = \text{span}\{b, Ab, \dots, A^{m-1}b\}.$$

Note that $\dim K_m(A; b) = \min(\deg_b(A), m)$, where $\deg_b(A)$ is the smallest possible degree of a polynomial p such that $p(A)b = 0$. In particular, $\dim K_m(A; b)$ is maximized when $\{b, Ab, \dots, A^{m-1}b\}$ are linearly independent.

(b)

Consider a square system. A Krylov subspace method seeks iterates of the form

$$x^{(k)} = x^{(0)} + q_{k-1}(A)r^{(0)},$$

where q_{k-1} is a polynomial of degree at most $k-1$. In particular, at iteration k , a Krylov subspace method seeks the best approximation of x in the set $x^{(0)} + K_k(A; r^{(0)}) := \{x^{(0)} + y : y \in K_k(A; r^{(0)})\}$.

The Jacobi and Gauss-Seidel methods are not Krylov subspace methods, but the conjugate gradient method is. Recall that the Jacobi and Gauss-Seidel methods are stationary, i.e., their iterations are of the form

$$x^{(k+1)} = x^{(k)} + P^{-1}r^{(k)},$$

where P is a fixed preconditioning matrix (for Jacobi, $P = D$, while for Gauss-Seidel, $P = D - E$). In contrast, the iterates of Krylov subspace methods look like

$$x^{(k+1)} = x^{(k)} + \sum_{j=0}^k \alpha_j r^{(j)}.$$

For Jacobi and Gauss-Seidel, the same transformation is applied to the residual at each step, unlike Krylov subspace methods.

However, the conjugate gradient method is a Krylov subspace method. Recall that CG computes a search direction $p^{(k)}$ as $p^{(0)} = r^{(0)}$ and for $k \geq 1$,

$$p^{(k)} = r^{(k)} - \beta_k p^{(k-1)}$$

for some scalar β_k . By induction, we see that $p^{(k)}$ is a linear combination of $r^{(0)}, \dots, r^{(k)}$. The iterates of CG are then

$$x^{(k+1)} = x^{(k)} + \alpha_k p^{(k)} = x^{(k)} + \sum_{j=0}^k \gamma_j r^{(j)}$$

for some scalars γ_j .

(c)

We claim that the power method on a matrix $A \in \mathbb{C}^{n \times n}$ converges for almost every choice of initial value (we will make this more precise soon) if A is diagonalizable and its maximal eigenvalue (more often called the *dominant eigenvalue*)

has algebraic multiplicity 1 and absolute value greater than all other eigenvalues.

Denote the eigenvalues of A by $\lambda_1, \dots, \lambda_n$ in descending order of absolute value (note that some λ_i may be the same, but $|\lambda_1| > |\lambda_i|$ for all $i = 2, \dots, n$ by hypothesis). Now, A being diagonalizable means we can take $x_1, \dots, x_n$ linearly independent unit eigenvectors of $\lambda_1, \dots, \lambda_n$ respectively. Hence, $x_1, \dots, x_n$ forms a basis of $\mathbb{C}^n$. Let $q^{(0)}$ be some initial guess and write $q^{(0)} = \sum_{j=1}^n \alpha_j x_j$. We now assume $\alpha_1 \neq 0$; although the α_j are not known at this stage, this assumption is not dangerous to make. This is because $\text{span}\{x_2, \dots, x_n\}$ has Lebesgue measure 0 in $\mathbb{C}^n$, so almost every choice of $q^{(0)}$ will result in $\alpha_1 \neq 0$. From here, define the iterates of the power method in the usual way:

$$\begin{aligned} z^{(k)} &= Aq^{(k-1)} \\ q^{(k)} &= z^{(k)} / \|z^{(k)}\|_2 \\ \nu^{(k)} &= (q^{(k)})^H Aq^{(k)}. \end{aligned}$$

Observe that $q^{(k)} = \frac{A^k q^{(0)}}{\|A^k q^{(0)}\|_2}$. Letting $\tilde{q}^{(k)} = \frac{q^{(k)} \|A^k q^{(0)}\|_2}{\alpha_1 \lambda_1^k}$, we have

$$\|\tilde{q}^{(k)} - x_1\|_2 = \left\| \sum_{j=2}^n \frac{\alpha_j}{\alpha_1} \left(\frac{\lambda_j}{\lambda_1} \right)^k x_j \right\|_2 \leq \left| \frac{\lambda_2}{\lambda_1} \right|^k \sum_{j=2}^n \left| \frac{\alpha_j}{\alpha_1} \right| \rightarrow 0$$

as $k \rightarrow \infty$ since $|\lambda_2| < |\lambda_1|$. Thus, $\tilde{q}^{(k)} \rightarrow x_1$, and so

$$\nu^{(k)} = (q^{(k)})^H Aq^{(k)} = \frac{(\tilde{q}^{(k)})^H A\tilde{q}^{(k)}}{\|\tilde{q}^{(k)}\|_2^2} \rightarrow x_1^H A x_1 = \lambda_1$$

since x_1 is a unit vector. So, the iterates $\nu^{(k)}$ converge to the dominant eigenvalue.

Problem 2

Consider $f(x) = 0$ and an iterative method $x_{n+1} = x_n + Af(x_n)$.

(a)

Define $\phi(x) = x + Af(x)$ and consider iterates $x_{n+1} = \phi(x_n)$. Provided that A is nonzero, we see that a fixed point x^* of ϕ satisfies $f(x^*) = 0$; conversely, any root of f is clearly a fixed point of ϕ . If such a point exists, then we can guarantee local convergence of these iterates if $\rho(J_\phi(x^*)) < 1$, where $J_\phi(x^*)$ denotes the Jacobian of ϕ at x^* . In this case, it is clear that $J_\phi = I + AJ_f$. So, the iterates have local convergence (i.e., if x_0 is sufficiently close to x^*) if $\rho(I + AJ_f(x^*)) < 1$.

Note that for any square matrix B , $1 + \lambda \in \sigma(I + B)$ if and only if $\lambda \in \sigma(B)$. Therefore, the spectrum of $I + AJ_f(x^*)$ being contained in $(-1, 1)$ is equivalent to the saying that the spectrum of $AJ_f(x^*)$ is contained in $(-2, 0)$.

(b)

Taking $A = -B^{-1}$ results in convergence in one step. For any initial point x_0 ,

$$x_1 = x_0 - B^{-1}(Bx_0 - c) = B^{-1}c,$$

and so

$$f(x_1) = B(B^{-1}c) - c = 0.$$

This choice of A is not practical since it reduces to solving $Bx = c$ directly.

(c)

Letting $A_n = -J_f^{-1}(x_n)$ results in the usual Newton's method for root-finding, which converges quadratically provided $f \in C^2$, $J_f(x^*)$ is nonsingular, and x_0 is sufficiently close to x^* .

Problem 3

(a)

Let $h = \frac{b-a}{3}$. Then, the quadrature nodes are $x_0 := a + h$ and $x_1 := b - h$. The Lagrange interpolating polynomial through x_0 and x_1 is therefore $\Pi_1 f(x) = f(x_0)\ell_0(x) + f(x_1)\ell_1(x)$, where $\ell_0(x) = \frac{x - x_1}{x_0 - x_1}$ and $\ell_1(x) = \frac{x - x_0}{x_1 - x_0}$ are the usual Lagrange basis polynomials. Now, the open Newton-Cotes formula is obtained by integrating this interpolant over $[a, b]$:

$$I_1(f) = \int_a^b \Pi_1 f(x) dx = f(x_0) \int_a^b \ell_0(x) dx + f(x_1) \int_a^b \ell_1(x) dx.$$

But now, the quadrature weights $\int_a^b \ell_i(x) dx$ can be written in a way that depends only on i and h , not $[a, b]$. Consider parameterizing x as $x = \Psi(t) := x_0 + th = a + (t+1)h$. Notice

$$\ell_0(x) = \frac{x - x_1}{x_0 - x_1} = \frac{(a + (t+1)h) - (a + 2h)}{a + h - (a + 2h)} = \frac{(t-1)h}{-h} = 1 - t.$$

Likewise,

$$\ell_1(x) = \frac{(a + (t+1)h) - (a + h)}{(a + 2h) - (a + h)} = \frac{th}{h} = t.$$

Now, $a = \Psi(-1)$ and $b = \Psi(2)$, so using a change of variables gives

$$\int_a^b \ell_0(x) dx = h \int_{-1}^2 (1 - t) dt = \frac{3}{2}h$$

and

$$\int_a^b \ell_1(x) dx = h \int_{-1}^2 t dt = \frac{3}{2}h.$$

(note $dx = h dt$ by the definition of Ψ). So, we can rewrite our formula as

$$I_1(f) = \frac{3}{2}h[f(x_0) + f(x_1)] = \frac{b-a}{2}[f(x_0) + f(x_1)].$$

Notice that this is just the trapezoidal rule, but the nodes are internal to (a, b) .

The midpoint rule is just an open Newton-Cotes formula with $n = 0$. In general, for any Newton-Cotes formula with $n + 1$ points, we have

$$\text{Degree of Exactness} = \begin{cases} n + 1 & n \text{ is even} \\ n & n \text{ is odd} \end{cases}$$

In particular, the open Newton-Cotes formulae with 1 or 2 nodes both have the same degree of exactness.

(b)

In general, a Gaussian quadrature rule with $n+1$ points has a degree of exactness of $2n+1$. In this case, using 2 points will yield a degree of exactness of 3, which is maximal.

(c)

For the Monte Carlo method, we sample points $x_1, \dots, x_n \in [a, b]$ uniformly (and independently) at random, and define the integral estimate as

$$I_n(f) = \frac{(b-a)}{n} \sum_{i=1}^n f(x_i).$$

(i.e., we estimate the “average value” by a sample mean, then multiply by the length of the interval). To determine an error bound, we consider iid random variables $X_1, \dots, X_n \sim \text{Uniform}(a, b)$. Notice that $I_{X_1, \dots, X_n}(f) := \frac{b-a}{n} \sum_{i=1}^n f(X_i)$ is unbiased for $I(f)$, since

$$\mathbb{E}[I_{X_1, \dots, X_n}(f)] = (b-a)\mathbb{E}[f(X_1)] = (b-a) \int_a^b \frac{1}{b-a} f(x) dx = \int_a^b f(x) dx.$$

Therefore, we can obtain an error estimate for $E_n(f) := I(f) - I_n(f)$ by considering the standard deviation:

$$(\mathbb{E}[(I(f) - I_{X_1, \dots, X_n}(f))^2])^{1/2} = (\text{Var}(I_{X_1, \dots, X_n}(f)))^{1/2} = \frac{b-a}{\sqrt{n}} (\text{Var}(f(X_1)))^{1/2}.$$

In particular, the error between $I(f)$ and the Monte Carlo integral estimate with n points is $O(n^{-1/2})$.

January 2024

Problem 1

(a)

Consider an overdetermined system and write the SVD of A by $A = U\Sigma V^T$; recall that $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ are orthogonal while $\Sigma \in \mathbb{R}^{m \times n}$ is rectangular with singular values on its diagonal in descending order. Since U is orthogonal, it preserves L^2 norms (cf. problem 1(a) of January 2023), so

$$\|Ax - b\|_2^2 = \|U(\Sigma V^T x - U^T b)\|_2^2 = \|\Sigma V^T x - U^T b\|_2^2.$$

Let r be the number of positive singular values of A and denote these singular values by $\sigma_1, \dots, \sigma_r$. Then, for any $w \in \mathbb{R}^n$,

$$\|\Sigma w - U^T b\|_2^2 = \sum_{i=1}^r (\sigma_i w_i - [U^T b]_i)^2 + \sum_{i=r+1}^m ([U^T b]_i)^2.$$

From here, define w^* by

$$w_i^* = \begin{cases} \frac{[U^T b]_i}{\sigma_i} & i = 1, \dots, r, \\ 0 & i = r+1, \dots, m. \end{cases}$$

We see that w^* minimizes $\|\Sigma w - U^T b\|_2^2$. Now, the desired least-squares solution x^* is given by $x^* = Vw^*$ since V is orthogonal.

Going a bit further, we realize that w^* is actually $w^* = \Sigma^\dagger U^T b$, and so

$$x^* = V\Sigma^\dagger U^T b = A^\dagger b,$$

where $\dagger$ denotes the Moore-Penrose pseudoinverse.

(b)

In an underdetermined system, there tend to be infinitely many solutions of the form $x = \hat{x} + v$, where $\hat{x}$ satisfies $A\hat{x} = b$ and $v \in \text{Null}(A)$. The solution $x^* = A^\dagger b$ from the Moore-Penrose pseudoinverse satisfies the following two conditions:

- (1) x^* minimizes $\|Ax - b\|_2$
- (2) x^* has the smallest L^2 norm out of any element of $\mathbb{R}^m$ that satisfies condition (1).

Consequently, x^* is also orthogonal to $\text{Null}(A)$. This is because x^* can be decomposed into $x^* = x_1 + x_2$ for $x_1 \in [\text{Null}(A)]^\perp$ and $x_2 \in \text{Null}(A)$. If $x_2 \neq 0$, then $\|x^*\|_2^2 = \|x_1\|_2^2 + \|x_2\|_2^2 > \|x_1\|_2^2$. However, $Ax^* = Ax_1$, meaning x_1 also satisfies condition (1), contradicting condition (2). Hence, $x^* = x_1$.

(c)

Given a vector $v \in \mathbb{R}^n$, the Householder transformation associated to v is the matrix

$$P_v = I_n - \frac{2vv^T}{\|v\|_2^2}.$$

Multiplying a vector $x \in \mathbb{R}^n$ by P_v amounts to reflecting x across the orthogonal complement of $\text{span}(v)$.

Householder transformations are used to reduce a square matrix to upper Hessenberg form. Let $A^{(0)} \in \mathbb{R}^{n \times n}$, and for $k = 1, \dots, n-2$, let

$$\begin{aligned} x^{(k)} &= k\text{th column of } A^{(k-1)} \\ w^{(k)} &= x_{n-k}^{(k)} + \|x_{n-k}^{(k)}\|_2 \mathbf{e}_1^{(n-k)} \\ P^{(k)} &= \begin{pmatrix} I_k & 0 \\ 0 & P_{w^{(k)}} \end{pmatrix} \\ A^{(k)} &= (P^{(k)})^T A^{(k-1)} P^{(k)} \end{aligned}$$

where $x_{n-k}^{(k)} \in \mathbb{R}^{n-k}$ is the vector formed from the last $n-k$ entries of $x^{(k)}$, and $\mathbf{e}_1^{(n-k)}$ is the first standard basis vector of $\mathbb{R}^{n-k}$. This similarity transformation results in all elements below the subdiagonal of $A^{(k)}$ being made zero in the first k columns. Thus, $A^{(n-2)}$ is in upper Hessenberg form.

Householder transformations are also used to transform a matrix $A \in \mathbb{R}^{m \times n}$ ($m \geq n$) into a bidiagonal matrix. This involves constructing sequences $U_1, \dots, U_{m-1}$ and $V_1, \dots, V_{n-2}$ of Householder matrices which are multiplied onto A (U_i are left-multiplied onto A while V_j are right-multiplied) to eliminate all elements below the main diagonal and above the superdiagonal. Hence, after $n+m-3$ multiplications, A is transformed into an upper bidiagonal matrix (i.e., all entries outside the diagonal and superdiagonal are zero).

Notice that multiplication by an orthogonal matrix does not necessarily preserve eigenvalues, but it does preserve singular values. In the first method, similarity transformations by Householder matrices are applied to A , which preserves eigenvalues. However, in the second method, Householder matrices are only multiplied onto A , not conjugated. Hence, even if A was square, the eigenvalues of A are not preserved, but the singular values are.

Problem 2

(a)

Consider a nonlinear system, and assume the Jacobian J_f exists on a neighborhood of a solution x^* to this system, and let $x^{(0)}$ be an initial point (usually,

$x^{(0)}$ is assumed to be sufficiently close to x^*). At each iteration $k \geq 0$, Newton's method first determines $\delta x^{(k)}$ such that

$$J_f(x^{(k)})\delta x^{(k)} = -f(x^{(k)}),$$

then generates a new iterate

$$x^{(k+1)} = x^{(k)} + \delta x^{(k)}.$$

(b)

In the scalar case ($n = 1$), Newton's method constructs a tangent line approximation of f at the current guess $x^{(k)}$. Provided $f'(x^{(k)}) \neq 0$, this tangent line will have a unique root; this root is the next iterate $x^{(k+1)}$.

(c)

We claim that Newton's method with $f'(x_n) = f'(\bar{x})$ fixed converges if $f \in C^1$ and $f'(x^*) \neq 0$, provided x_0 and $\bar{x}$ are sufficiently close to x^* (we will specify how close in the proof).

Proving convergence for methods like these usually comes down to casting it as a fixed-point iteration. Define

$$\phi(x) = x - \frac{f(x)}{f'(\bar{x})},$$

for $\bar{x}$ soon to be determined. The idea is to force $|\phi'| \leq C < 1$ on a neighborhood around x^* . The continuity of f' and the fact that $f'(x^*) \neq 0$ means that f' is nonzero on some neighborhood $B_r(x^*)$ of x^* . Furthermore, the continuity of f' means we can take $\bar{x} \in B_r(x^*)$ such that

$$\frac{1}{2} < \frac{f'(x^*)}{f'(\bar{x})} < \frac{3}{2}.$$

With this, ϕ is well-defined and C^1 on $B_r(x^*)$, with

$$\phi'(x) = 1 - \frac{f'(x)}{f'(\bar{x})}.$$

Hence,

$$|\phi'(x^*)| = \left| 1 - \frac{f'(x^*)}{f'(\bar{x})} \right| < \frac{1}{2}.$$

It follows that there is another neighborhood $B_s(x^*) \subseteq B_r(x^*)$ on which $|\phi'| \leq C < 1$. Let $x_0 \in B_s(x^*)$, and notice that the Newton iterates are the fixed-point iterates $x_{n+1} := \phi(x_n)$. Then, for some ξ_0 between x^* and x_0 ,

$$\begin{aligned} |x_1 - x^*| &= |\phi(x_0) - \phi(x^*)| && \text{since } \phi(x^*) = x^* \\ &= |\phi'(\xi_0)| \cdot |x_0 - x^*| && \text{mean value theorem} \\ &\leq C |x_0 - x^*| && \text{since } \xi_0 \in B_s(x^*). \end{aligned}$$

Hence, $x_1 \in B_s(x^*)$. Proceeding by induction will immediately reveal that $x_n \in B_s(x^*)$ for all n , with

$$|x_n - x^*| \leq C^n |x_0 - x^*| \leq sC^n.$$

Since $0 < C < 1$, we have $x_n \rightarrow x^*$.

Problem 3

(a)

The trapezoidal rule is obtained by integrating a linear Lagrange interpolant. In particular, we consider the Lagrange interpolant with nodes at a and b , which is

$$\Pi_1 f(x) = f(a) \cdot \frac{x-b}{a-b} + f(b) \cdot \frac{x-a}{b-a}.$$

Integrating on $[a, b]$ yields

$$\begin{aligned} I_1(f) &= \int_a^b \Pi_1 f(x) dx = \frac{f(a)}{b-a} \left[bx - \frac{x^2}{2} \right] \Big|_{x=a}^b + \frac{f(b)}{b-a} \left[\frac{x^2}{2} - ax \right] \Big|_{x=a}^b \\ &= \frac{f(a)}{b-a} \left[\frac{b^2}{2} - ab + \frac{a^2}{2} \right] + \frac{f(b)}{b-a} \left[\frac{b^2}{2} - ab + \frac{a^2}{2} \right] \\ &= \frac{(b-a)^2}{2(b-a)} [f(a) + f(b)] \\ &= \frac{b-a}{2} [f(a) + f(b)], \end{aligned}$$

which is the trapezoidal rule.

The composite trapezoidal rule is obtained by integrating a piecewise linear interpolant, which amounts to summing up trapezoidal rules on each subinterval. Specifically, if $x_0 < \dots < x_n$ are the interpolation nodes and $\Pi_{1,n}(f)$ is the piecewise linear interpolant of f through these nodes (notice that $\Pi_{1,n}f$ has n components), then the composite trapezoidal rule is

$$I_{1,n}(f) = \int_a^b \Pi_{1,n} f(x) dx = \sum_{i=1}^n \frac{x_i - x_{i-1}}{2} [f(x_i) + f(x_{i-1})].$$

Now, if $x_i - x_{i-1} = h$ for $i = 1, \dots, n$ (i.e., the nodes are equidistant), the composite trapezoidal rule reads

$$I_h^1(f) = \frac{h}{2} \sum_{i=1}^n [f(x_i) + f(x_{i-1})] = h \left[\frac{1}{2} f(x_0) + f(x_1) + \dots + f(x_{n-1}) + \frac{1}{2} f(x_n) \right].$$

(b)

Assuming $f \in C^2(a, b)$, the error for linear interpolation is given by

$$f(x) - \Pi_1 f(x) = \frac{f''(\xi_x)}{2}(x-a)(x-b),$$

for $\xi_x \in (a, x)$ dependent on x . Integrating this along $[a, b]$ yields

$$E_1(f) = \frac{1}{2} \int_a^b f''(\xi_x)(x-a)(x-b) dx = -\frac{1}{2} \int_a^b f''(\xi_x)(x-a)(b-x) dx.$$

When we write the integrand in this way, we see that $(x-a)(b-x)$ is nonnegative on $[a, b]$. The so-called mean value theorem for integrals tells us we can take some $\xi \in (a, b)$ such that

$$\begin{aligned} -\frac{1}{2} \int_a^b f''(\xi_x)(x-a)(b-x) dx &= -\frac{f''(\xi)}{2} \int_a^b (x-a)(b-x) dx \\ &= -\frac{f''(\xi)}{2} \int_0^{b-a} u(b-a-u) du \quad (u = x-a) \\ &= -\frac{f''(\xi)}{2} \left[-\frac{u^3}{3} + \frac{(b-a)u^2}{2} \right] \Big|_{x=0}^{b-a} \\ &= -\frac{f''(\xi)}{12} (b-a)^3. \end{aligned}$$

In particular, if f'' is bounded on (a, b) ,

$$|E_1(f)| \leq \frac{\|f''\|_\infty}{12} (b-a)^3.$$

(c)

Altering an interpolation value on a linear B -spline interpolant affects at most 2 linear components, namely, the components whose endpoints are at the altered value. Thus, the interpolant is affected on at most two subintervals.

With quadratic B -splines, the fact that the interpolant must be C^1 means that altering one interpolation value could in principle affect all quadratics. Intuitively, changing one interpolation value will clearly affect the two quadratics whose endpoints are at that value, and changing these two quadratics will then force their neighboring quadratics to change in order to enforce differentiability, and so on.

August 2023

Problem 1

A is symmetric, strictly positive definite.

(a)

Strictly positive definite $\iff$ all eigenvalues are positive $\iff$ trivial null space $\iff$ nonsingular.

(b)

Let n be the number of rows of A . The Cholesky decomposition of A is of the form $A = H^T H$, where H^T is a lower triangular matrix defined in the following way:

Let $h_{11} = \sqrt{a_{11}}$ (every diagonal entry of A is strictly positive, so this can be done). Then, for $i = 2, \dots, n$,

$$h_{ij} = \left(a_{ij} - \sum_{k=1}^{j-1} h_{ik} h_{jk} \right) / h_{jj}, \quad j = 1, \dots, i-1$$

$$h_{ii} = \left(a_{ii} - \sum_{k=1}^{i-1} h_{ik}^2 \right)^{1/2}.$$

In the i th row, each element h_{ij} requires j multiplications, of which there are $i-1$ such terms. In addition, element h_{ii} requires $i-1$ multiplications. Thus, the number of multiplications to form row i of H is

$$i-1 + \sum_{j=1}^{i-1} j = \left(\sum_{j=1}^i j \right) - 1 = \frac{i(i+1)}{2} - 1.$$

So, the total number of multiplications to form H is

$$\begin{aligned} \sum_{i=2}^n \left(\frac{i(i+1)}{2} - 1 \right) &= \frac{1}{2} \left(\sum_{i=1}^n i^2 + i - 2 \right) = \frac{1}{2} \left(\frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} - 2n \right) \\ &= \frac{1}{6} (n^3 + 3n^2 - 4n). \end{aligned}$$

In particular, forming the matrix H is $O(n^3)$.

(c)

The pseudoinverse of A is

$$x^* = A^\dagger b = V \Sigma^\dagger U^T b,$$

where $A = U \Sigma V^T$ is an SVD of A , and

$$\Sigma^\dagger = \text{diag}(\sigma_1^{-1}, \dots, \sigma_r^{-1}, 0, \dots, 0)$$

for $\sigma_1, \dots, \sigma_r$ the nonzero singular values of A (in descending order). In January 2024, problem 1(a/b), we saw how it can solve over/underdetermined systems and properties of the solution.

Problem 2

(a)

Suppose $f : \mathbb{R}^d \rightarrow \mathbb{R}$ has a unique minimum x^* . Newton's method for minimization is a particular kind of descent method. We start with a point $x^{(0)} \in \mathbb{R}^d$, then at step $k \geq 0$, take $d^{(k)}$ that solves

$$H_f(x^{(k)})d^{(k)} = -\nabla f(x^{(k)}),$$

where H_f is the Hessian of f and ∇f is the gradient of f . Then, put

$$x^{(k+1)} = x^{(k)} + d^{(k)}.$$

(b)

The third derivative condition just ensures that x^* is a global minimum of f'' , and thus f is ε -strongly convex with unique global minimum at x^* . Since f is ε -strongly convex, x^* is the sole root of f' . For $d = 1$, the Newton iterates are

$$x^{(k+1)} = x^{(k)} - \frac{f'(x^{(k)})}{f''(x^{(k)})}.$$

We see that the above iterates are nothing but the iterates for Newton's method for solving $f'(x) = 0$. So, any proof of convergence for Newton's method for solving scalar equations applied to f' can be used here. The proof in August 2019, problem 2a will suffice, as will using fixed-point iterations.

(c)

The constraint $0 \leq x \leq 1$ is equivalent to

$$\max(x - 1, 0) = \min(x, 0) = 0.$$

Then, for a sequence (α_k) of increasing, positive scalars tending to $+\infty$, we solve the unconstrained minimization problem

$$\text{minimize } \left(f(x) + \frac{\alpha_k}{2} [\max(x - 1, 0)^2 + \min(x, 0)^2] \right)$$

for all k . The solutions x_k^* to the k -th minimization problem, assuming they exist, converge to a constrained minimum of f on $[0, 1]$ if f is continuous on $[0, 1]$.

Problem 3

(a)

We claim that the midpoint rule is the Gauss quadrature rule with 1 point. We first assume $[a, b] = [-1, 1]$, then will generalize. For a Gaussian quadrature

rule using 1 point, this node is given by the sole root of the first Legendre polynomial. This polynomial is simply $L_1(x) = x$, and so the node is at 0. The corresponding node in a general interval $[a, b]$ is found by applying the affine transformation

$$\psi(x) = \frac{b-a}{2}x + \frac{a+b}{2}$$

to each node. In particular, $\psi(0) = \frac{a+b}{2}$, and so the quadrature node is at the midpoint. The interpolating polynomial through this point is the constant polynomial $\Pi_0 f(x) \equiv f((a+b)/2)$, and so the Gaussian quadrature rule is obtained simply by integrating this interpolant:

$$I_0(f) = \int_a^b \Pi_0 f(x) dx = (b-a)f((a+b)/2),$$

which is the midpoint rule.

(b)

Assume $f \in C^4$, and let $c = (a+b)/2$. Writing the Taylor expansion of f about c yields

$$f(x) = f(c) + f'(c)(x-c) + \frac{f''(c)(x-c)^2}{2} + \frac{f'''(c)(x-c)^3}{6} + \frac{f^{(4)}(\xi_x)(x-c)^4}{24}$$

for some $\xi_x \in (a, b)$ dependent on x . Notice that for $k \in \mathbb{N}$,

$$\int_a^b (x-c)^k dx = \int_{a-c}^{b-c} u^k du = \frac{1}{k+1} u^{k+1} \Big|_{(a-b)/2}^{(b-a)/2} = \begin{cases} 0 & k \text{ is odd} \\ \frac{2}{k+1} \left(\frac{b-a}{2}\right)^{k+1} & k \text{ is even} \end{cases}.$$

So, integrating both sides of the Taylor expansion yields

$$I(f) = (b-a)f(c) + \frac{f''(c)}{24}(b-a)^3 + \frac{1}{24} \int_a^b f^{(4)}(\xi_x)(x-c)^4 dx.$$

Notice that the first term on the right is the midpoint formula, while the third term on the right is of order h^5 for $h := b-a$. Hence,

$$E_0(f) = I(f) - I_0(f) = \frac{f''(c)}{24}h^3 + O(h^5).$$

NB: This is not the standard way to obtain an error estimate for the midpoint rule, since the standard error estimate is not written in terms of $f''(c)$. The standard way is to expand only up to order 2, then integrating. With that estimate, there is only an assumption of C^2 that is needed. Nonetheless, the way I have it written here is what I believe the problem is asking for.

(c)

January 2025, problem 3(c).

January 2023

Problem 1

(a)

Let B be an invertible matrix such that $B^{-1}AB$ is defined. If $Ax = \lambda x$ for some $x \neq 0$, then

$$B^{-1}AB(B^{-1}x) = B^{-1}Ax = \lambda(B^{-1}x).$$

B being nonsingular means $B^{-1}x \neq 0$, so λ is also an eigenvalue of $B^{-1}AB$. Likewise, suppose $B^{-1}AB y = \mu y$ for $y \neq 0$. Then,

$$A(By) = B(\mu y) = \mu(By),$$

and since $By \neq 0$, μ is also an eigenvalue of A .

Next, let A be a matrix and U unitary. Recall that the L^2 norm of a matrix is equivalent to its largest singular value, i.e., the square root of the largest eigenvalue of $A^H A$. But now,

$$\begin{aligned}\|UA\|_2^2 &= \text{largest eigenvalue of } (UA)^H UA \\ &= \text{largest eigenvalue of } A^H U^H U A \\ &= \text{largest eigenvalue of } A^H A \\ &= \|A\|_2^2.\end{aligned}$$

Likewise,

$$\begin{aligned}\|AU\|_2^2 &= \text{largest eigenvalue of } (AU)^H AU \\ &= \text{largest eigenvalue of } U^{-1}(A^H A)U \\ &= \text{largest eigenvalue of } A^H A \\ &= \|A\|_2^2,\end{aligned}$$

since, from the first part of this problem, similarity transformations preserve the spectrum of A .

If, instead, the problem means that unitary transformations preserve the L^2 norm of a *vector*, let x be a column vector conformable to multiplication with U , and observe

$$\|Ux\|_2^2 = (Ux)^H (Ux) = x^H x = \|x\|_2^2.$$

(b)

Taking $1 \leq i \leq k \leq n$ and $\theta \in [0, 2\pi)$, the Givens rotation between the i th and k th coordinates of $\mathbb{R}^n$ by θ is defined as $G(i, k, \theta) := I_n - Y$, where Y is an $n \times n$ matrix with $y_{ii} = y_{kk} = \cos(\theta)$, $y_{ki} = \sin(\theta) = -y_{ik}$, and zeros in all other components.

In the QR algorithm, Givens rotations are used for computing a QR factorization of a matrix in upper Hessenberg form. Let $H^{(0)}$ be one such matrix in upper Hessenberg form. For $k \in \mathbb{N}$ and $j = 1, \dots, n-1$, define $G_j^{(k)} = G(j, j+1, \theta_j)^{(k)}$, where θ_j is chosen so that the $(j+1, j)$ -th element of $(G_1^{(k)} \dots G_j^{(k)})^T H^{(k-1)}$ is zero. Then, let

$$\begin{aligned} Q^{(k)} &= G_1^{(k)} \dots G_{n-1}^{(k)} \\ R^{(k)} &= (Q^{(k)})^T H^{(k-1)}, \\ H^{(k)} &= R^{(k)} Q^{(k)}. \end{aligned}$$

It is a fact that Givens matrices are orthogonal, so $Q^{(k)}$ is orthogonal for any k . What's more, each $Q^{(k)}$ is in upper Hessenberg form, and $R^{(k)}$ is upper triangular by the construction of the Givens rotations (each Givens rotation eliminates a subdiagonal entry of $H^{(k-1)}$). So, $H^{(k-1)} = Q^{(k)} R^{(k)}$ is a QR factorization, and $H^{(k)}$ is also in upper Hessenberg form, being a product of an upper triangular matrix and a matrix in upper Hessenberg form. Hence, by induction, $H^{(k)}$ is in upper Hessenberg form for all k , which allows us to use Givens rotations at each step to generate a QR decomposition.

We see that

$$H^{(k)} = (Q^{(k)})^T H^{(k-1)} Q^{(k)}$$

for all k , and so by induction,

$$H^{(k)} = (Q^{(1)} Q^{(2)} \dots Q^{(k)})^T H^{(0)} Q^{(1)} Q^{(2)} \dots Q^{(k)}.$$

It is known that as $k \rightarrow \infty$, $H^{(k)}$ converges to a matrix whose eigenvalues can be immediately read off (in particular, eigenvalues are found in diagonal blocks of order no more than 2). Since $H^{(k)}$ is similar to $H^{(0)}$ for all k , this complete set of eigenvalues must be the eigenvalues of $H^{(0)}$ by part (a).

(c)

As seen in part (b), the QR process results in a similarity transformation by an orthogonal matrix. By part (a), similarity transformations by orthogonal matrices preserve eigenvalues and the L^2 norm.

Problem 2

(a)

Let f denote the function we seek to minimize. Note that

$$\nabla f(x) = \begin{pmatrix} 2(x_1 + 1) \\ 2(x_2 - 1) \end{pmatrix}, \quad H_f(x) = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 2I.$$

Starting from $x^{(0)} \in \mathbb{R}^2$, the Newton iterates are

$$x^{(k+1)} = x^{(k)} - H_f^{-1}(x^{(k)})\nabla f(x^{(k)}) = x^{(k)} - \begin{pmatrix} x_1^{(k)} + 1 \\ x_2^{(k)} - 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

Note that the right hand side is the unconstrained minimum of f . Indeed, since f is quadratic, Newton's method converges in one step.

(b)

We have a sequence of minimization problems of the form

$$\text{minimize } \left(f(x) + \frac{\alpha_k}{2} \min(x_1, 0)^2 \right)$$

Note that the gradient of this penalized objective function f_k is

$$\nabla f_k(x) = \begin{pmatrix} 2(x_1 + 1) + \alpha_k \min(x_1, 0) \\ 2(x_2 - 1) \end{pmatrix}.$$

On the other hand, if $x = (x_1, x_2)$ such that $x_1 < 0$, then

$$H_{f_k}(x) = \begin{pmatrix} 2 + \alpha_k & 0 \\ 0 & 2 \end{pmatrix},$$

and if $x_1 > 0$,

$$H_{f_k}(x) = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 2I.$$

If $x_1 = 0$, the Hessian is not defined since $\min(x_1, 0)$ is not differentiable with respect to x_1 at 0. Provided this line of non-differentiability is avoided, Newton's method can still be employed to minimize f_k for all k . In fact, we find analytically that the minimum of f_k is

$$x_k^* = \left(-\frac{2}{2 + \alpha_k}, 1 \right)$$

for all k . Notice that $-\frac{2}{2 + \alpha_k} \neq 0$, so f is C^2 on a neighborhood around x_k^* with an invertible Hessian. Hence, Newton's method on f_k will locally converge to x_k^* , and since f and $\min(x_1, 0)$ are both continuous, the sequence x_k^* converges to the constrained minimizer.

(c)

We have one constraint of $g(x_1, x_2) := -x_1 \leq 0$, so the Lagrangian is

$$\mathcal{L}(x_1, x_2, \lambda) = (x_1 + 1)^2 + (x_2 - 1)^2 - \lambda x_1.$$

If a minimum $x^* = (x_1^*, x_2^*)$ exists, there must exist some $\lambda^* \in \mathbb{R}$ such that the following four Kuhn-Tucker conditions are satisfied:

1. $x_1^* \geq 0$
2. $\lambda^* \geq 0$
3. $\nabla_x \mathcal{L}(x_1^*, x_2^*, \lambda^*) = 0$
4. $\lambda^* (-x_1^*) = 0$

Now,

$$\nabla_x \mathcal{L}(x_1^*, x_2^*, \lambda^*) = \begin{pmatrix} 2(x_1^* + 1) - \lambda^* \\ 2(x_2^* - 1) \end{pmatrix},$$

and condition (3) forces $x_2^* = 1$ and $\lambda^* = 2(x_1^* + 1)$. On the other hand, condition (4) forces $\lambda^* = 0$ or $x_1^* = 0$. If $\lambda^* = 0$, then $x_1^* = -1$, contradicting condition (1). If $x_1^* = 0$, we obtain $\lambda^* = 2$, and the triple $(x_1^*, x_2^*, \lambda^*) = (0, 1, 2)$ satisfies all four necessary Kuhn-Tucker conditions. Since f and the constraint g are convex, it follows that $(0, 1)$ is in fact the constrained minimum.

Problem 3

(a)

January 2025, problem 3(a) with $a = 0$ and $b = 3$.

(b)

The order is h^3 , provided $f \in C^2$ (remember that Newton-Cotes with 2 points is a type of trapezoidal rule).

If we instead use a Gaussian quadrature rule (i.e., the quadrature nodes are found by taking the roots of the second Legendre polynomial, then rescaling to $[0, 3]$), the order of accuracy increases to 5; however, this fact assumes $f \in C^4$.

(c)

The resulting quadrature formula is just Simpson's rule, since the interpolatory nodes being used are the endpoints and the midpoint:

$$I_2(f) = \frac{1}{2}[f(0) + 4f(1.5) + f(3)].$$

August 2020

The exam file for these problems does say August 2020, but on the UT prelim website, it's listed under August 2021 *and* August 2020.

Problem 1

(a)

See January 2024, problem 1(c) for the method of transforming a matrix to upper Hessenberg form using Householder transformations.

Since Householder matrices are orthogonal (they are symmetric and equal to their own inverse, in fact), it follows from induction that $A^{(n-2)}$ is similar to $A^{(0)}$, so they share the same set of eigenvalues (January 2023, problem 1(a)). Thus, the QR algorithm can instead be applied to $A^{(n-2)}$ to find the complete set of eigenvalues of $A^{(0)}$; the benefit of running the QR algorithm on $A^{(n-2)}$ is that Givens rotations can be used to reduce the computational cost of forming the QR factorizations.

(b)

Given $A \in \mathbb{R}^{n \times n}$ and $Q^{(0)} \in \mathbb{R}^{n \times n}$ some orthogonal matrix, the QR algorithm takes $A^{(0)} = (Q^{(0)})^T A Q^{(0)}$, and for $k \geq 1$,

Compute a QR decomposition $Q^{(k)} R^{(k)}$ of $A^{(k-1)}$
Set $A^{(k)} = R^{(k)} Q^{(k)}$.

In particular, $A^{(k)} = (Q^{(k)})^T A^{(k-1)} Q^{(k)}$, so by induction,

$$A^{(k)} = (Q^{(1)} \dots Q^{(k)})^T A^{(0)} (Q^{(1)} \dots Q^{(k)}).$$

Since $Q^{(0)} \dots Q^{(k)}$ is orthogonal, we have that $A^{(k)}$ is similar to A for all k . In particular, the limit of this sequence $A^{(k)}$ is upper triangular and similar to $A^{(0)}$, so the eigenvalues of A are simply the diagonal entries of $A^{(k)}$, provided A has real eigenvalues all distinct in absolute value.

(c)

Consider an overdetermined system, and assume A is full rank. Let $A = QR$ be a QR decomposition of A ; recall that $Q \in \mathbb{R}^{m \times m}$ is orthogonal and $R \in \mathbb{R}^{m \times n}$ is upper trapezoidal. Orthogonal matrices preserve the L^2 norm, so

$$\|Ax - b\|_2^2 = \|QRx - b\|_2^2 = \|Q(Rx - Q^T b)\|_2^2 = \|Rx - Q^T b\|_2^2.$$

From here, define $\tilde{Q} = Q(1 : m, 1 : n)$ (i.e., take the first n columns), and $\tilde{R} = R(1 : n, 1 : n)$ (i.e., take the first n rows). Because R is upper trapezoidal,

we have

$$\|Rx - Q^T b\|_2^2 = \|\tilde{R}x - \tilde{Q}^T b\|_2^2 + \sum_{i=n+1}^m ([Q^T b]_i)^2.$$

The sum on the right is constant in x , so the least-squares solution is attained by the minimizer $x^* := \tilde{R}^{-1}\tilde{Q}^T b$; the fact that A is full rank implies that $\tilde{R}$ is invertible since $\tilde{R}$ is actually the Cholesky factor of $A^T A$.

Problem 2

(a)

August 2023, problem 2(a)

(b)

August 2023, problem 2(b)

(c)

Let $B \in \mathbb{R}^{m \times d}$ and $b \in \mathbb{R}^m$. Assuming that “ $Bx - b \leq 0$ ” means “each component of $Bx - b$ is nonnegative”, the penalized objective function is

$$f_\alpha(x) = f(x) + \alpha \sum_{i=1}^m \max((Bx - b)_i, 0)^2.$$

For a sequence (α_k) of positive, increasing scalars tending to $+\infty$, we seek to solve

$$\text{minimize } f_{\alpha_k}(x), \quad x \in \mathbb{R}^d$$

for each k , and the solutions x_k^* to each problem converges to a solution to the constrained problem.

Problem 3

(a)

August 2023, problem 3(b) gives one way to do this. Another way is to expand only up to the second order. If $f \in C^2$, then for $x \in (-h, h)$,

$$f(x) = f(0) + f'(0)x + \frac{f''(\xi_x)}{2}x^2$$

for ξ_x between 0 and x . Integrating this along $(-h, h)$ yields

$$I(f) = 2hf(0) + \frac{1}{2} \int_{-h}^h f''(\xi_x)x^2 dx.$$

Now, the mean value theorem for integrals tells us that there exists a single $\xi \in (-h, h)$ such that

$$\frac{1}{2} \int_{-h}^h f''(\xi_x) x^2 dx = \frac{f''(\xi)}{2} \int_{-h}^h x^2 dx = \frac{h^3}{3} f''(\xi),$$

so the error can be written as

$$E_0(f) = I(f) - 2hf(0) = \frac{h^3}{3} f''(\xi).$$

In particular,

$$|E_0(f)| \leq \frac{h^3}{3} \|f''\|_\infty$$

(b)

Assume f is analytic and only consider one interval with length $h = b - a$. Expanding f in a Taylor series around $c := (b + a)/2$ gives

$$f(x) = f(c) + \sum_{k=1}^{\infty} \frac{f^{(k)}(c)}{k!} (x - c)^k.$$

Assuming we can integrate under the summation, we obtain

$$\begin{aligned} I(f) &= hf(c) + \sum_{k=1}^{\infty} \frac{f^{(k)}(c)}{k!} \int_a^b (x - c)^k dx \\ &= I_0(f) + \sum_{k=1}^{\infty} \frac{f^{(k)}(c)}{k!} \int_{(a-b)/2}^{(b-a)/2} u^k du \\ &= I_0(f) + \sum_{k=1}^{\infty} \frac{f^{(2k)}(c)}{(2k)!} \int_{(a-b)/2}^{(b-a)/2} u^{2k} du \\ &= I_0(f) + \sum_{k=1}^{\infty} \frac{f^{(2k)}(c)}{2^{2k}(2k+1)(2k)!} (b-a)^{2k+1} \\ &= I_0(f) + \sum_{k=1}^{\infty} C_k h^{2k} (hf^{(2k)}(c)). \end{aligned}$$

From here, recognize that $hf^{(2k)}(c)$ is a midpoint rule applied to $f^{(2k)}$. In particular,

$$hf^{(2k)}(c) = \int_a^b f^{(2k)}(x) dx + O(h^3) = f^{(2k-1)}(b) - f^{(2k-1)}(a) + O(h^3).$$

Now, consider nodes $a = x_0 < \dots < x_n = b$, where $x_{i+1} = x_i + h$ and $m_i = x_i + \frac{h}{2}$ are the midpoints. We can consider $I(f)$ as the sum of the integrals of f across

each subinterval. On each subinterval, we consider the Taylor expansion of f around m_i :

$$\begin{aligned}
I(f) &= \sum_{i=0}^{n-1} \int_{x_i}^{x_{i+1}} f(x) dx \\
&= h \sum_{i=0}^{n-1} f(m_i) + \sum_{i=0}^{n-1} \sum_{k=1}^{\infty} \frac{f^{(k)}(m_i)}{k!} \int_{x_i}^{x_{i+1}} (x - m_i)^k dx \\
&= I_{0,n}(f) + \sum_{k=1}^{\infty} C_k \sum_{i=0}^{n-1} h f^{(2k)}(m_i) (x_{i+1} - x_i)^{2k} \\
&= I_{0,n}(f) + \sum_{k=1}^{\infty} C_k h^{2k} \sum_{i=0}^{n-1} h f^{(2k)}(m_i).
\end{aligned}$$

By recognizing that $\sum_{i=0}^{n-1} h f^{(2k)}(m_i)$ is a composite midpoint rule for $f^{(2k)}$, we have the asymptotic expansion

$$I_{0,n}(f) = I(f) - \sum_{k=1}^{\infty} C_k h^{2k} [f^{(2k-1)}(b) - f^{(2k-1)}(a)].$$

(c)

The method here is very similar to Romberg integration. By applying Richardson extrapolation to the composite midpoint rule, we can eliminate lower order terms and improve the accuracy of this method. First, let $\mathcal{A}(h)$ denote the composite midpoint rule with subintervals of length h , and fix $\delta \in (0, 1)$, $h > 0$, and $n \in \mathbb{N}$. In the asymptotic expansion from part (b), only even-indexed terms survived, so the Richardson extrapolation algorithm has iterates of the following form:

$$\begin{aligned}
\mathcal{A}_{m,0} &= \mathcal{A}(\delta^m h), & m &= 0, \dots, n, \\
\mathcal{A}_{m,q+1} &= \frac{\mathcal{A}_{m,q} - \delta^{2(q+1)} \mathcal{A}_{m-1,q}}{1 - \delta^{2(q+1)}} & q &= 0, \dots, n-1, \quad m = q+1, \dots, n
\end{aligned}$$

Since the subinterval lengths $\delta^m h$ must divide the interval length $b-a$ an integer number of times, one often takes $h = b-a$ and $\delta = \frac{1}{2}$. With this, the iterates look like

$$\begin{aligned}
\mathcal{A}_{m,0} &= \mathcal{A}(2^{-m}(b-a)), & m &= 0, \dots, n, \\
\mathcal{A}_{m,q+1} &= \frac{4^{q+1} \mathcal{A}_{m,q} - \mathcal{A}_{m-1,q}}{4^{q+1} - 1} & q &= 0, \dots, n-1, \quad m = q+1, \dots, n
\end{aligned}$$

Note that $\mathcal{A}_{m,0}$ is simply the composite midpoint rule with 2^m subintervals of equal length.

January 2020

The exam file for these problems does say January 2020, but on the UT prelim website, it's listed under January 2022.

Problem 1

(a)

Consider a square system with A having all nonzero diagonal entries. The iterations of the Jacobi method begin with an arbitrary $x^{(0)} \in \mathbb{R}^n$, and for $k \in \mathbb{N}$,

$$x_i^{(k+1)} = a_{ii}^{-1} \left(b_i - \sum_{\substack{j=1 \\ j \neq i}}^n a_{ij} x_j^{(k)} \right), \quad i = 1, \dots, n.$$

This can also be formulated as

$$x^{(k+1)} = (I - D^{-1}A)x^{(k)} + D^{-1}b,$$

where D is the diagonal matrix consisting of the diagonal entries of A .

The Jacobi method (and other stationary methods) converge if and only if the spectral radius of the iteration matrix is less than 1. In this case, if $A = \begin{pmatrix} 2 & 4 \\ 1 & 1 \end{pmatrix}$, then $I - D^{-1}A = \begin{pmatrix} 0 & -2 \\ -1 & 0 \end{pmatrix}$. Its eigenvalues are $\pm\sqrt{2}$, both of which have absolute value greater than 1. So, the Jacobi method will not converge for this choice of A .

(b)

For $a, b \neq 0$, let $P = \begin{pmatrix} a^{-1} & 0 \\ 0 & b^{-1} \end{pmatrix}$ be some invertible diagonal matrix (we preemptively write a^{-1} and b^{-1} here for notational simplicity). So, we now consider the preconditioned system $P^{-1}Ax = P^{-1}b$, and $P^{-1}A = \begin{pmatrix} 2a & 4a \\ b & b \end{pmatrix}$. Letting D_P be the diagonal matrix of $P^{-1}A$, the Jacobi method's iteration matrix is now

$$B_P = I - D_P^{-1}P^{-1}A = \begin{pmatrix} 0 & -2 \\ -1 & 0 \end{pmatrix}.$$

Here, the eigenvalues are still ± 2 , so a diagonally preconditioned Jacobi method still does not converge.

Next, let $U = \begin{pmatrix} 2 & 4 \\ 0 & 1 \end{pmatrix}$ be the upper triangular part of A . This time, if D_U is the diagonal matrix of $Q := U^{-1}A$, the iteration matrix is

$$B_U = I - D_Q^{-1}Q^{-1}A = \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix}.$$

Since the iteration matrix is nilpotent (in particular, $B^2 = 0$), preconditioning with U will make the Jacobi method converge in 2 steps.

(c)

The power method converges to the largest eigenvalue (largest in the sense of having the largest absolute value; this is typically called the *dominant eigenvalue* instead) provided that the matrix is diagonalizable and the dominant eigenvalue has algebraic multiplicity 1 and an absolute value greater than all other eigenvalues. In this case, we find that the eigenvalues of A are $\frac{3 \pm \sqrt{17}}{2}$. Since A has two distinct eigenvalues, it is diagonalizable. Here, $\frac{3 + \sqrt{17}}{2}$ is the dominant eigenvalue and has algebraic multiplicity 1 and a greater absolute value than the other eigenvalue. Hence, the power method will converge to this eigenvalue and its associated unit eigenvector.

Inverse iteration can also be used to find the dominant eigenvalue, provided that the shift parameter μ is closer to $\frac{3 + \sqrt{17}}{2}$. Taking $\mu = 3$ will suffice for this end.

Problem 2

(a)

Take $h(x, y) = \begin{pmatrix} f(x, y) \\ g(x, y) \end{pmatrix}$ and write the same Newton's method iterates as January 2024, problem 2(a) for h .

Obviously, f and g have to depend on each other in some way, since if they don't have a root at the same point, there is no solution to $f(x, y) = g(x, y) = 0$. Moreover, if such a root $p^* = (x^*, y^*)$ did exist, then for the Newton iterations to be well-defined, we would also require $J_h(p^*)$ to be invertible, i.e., $f_x(p^*)g_y(p^*) - f_y(p^*)g_x(p^*) \neq 0$.

(b)

We will scrap the notation from part (a) and just consider a system $f(x) = 0$ for $x \in \mathbb{R}^2$ (this will actually work with any number of dimensions). Suppose a solution exists, which we will denote by x^* . There are a few assumptions we

will need to get quadratic convergence. First, assume f is C^1 on some open ball around x^* , that $J_f(x^*)$ is invertible, and there are positive constants R, C, L such that $\|J_f^{-1}(x^*)\| \leq C$ and

$$\|J_f(x) - J_f(y)\| \leq L \|x - y\| \quad x, y \in B_R(x^*).$$

Here, the norm $\|\cdot\|$ on the left hand side can be any consistent matrix norm, e.g., the Frobenius norm.

By the inverse function theorem, there is some ball $B_s(x^*)$ on which J_f is invertible. Let $r = \min(s, R, (2CL)^{-1})$. If $x^{(0)} \in B_r(x^*)$, then $(J_f(x^{(0)}))^{-1}$ exists, so the next iterate $x^{(1)}$ is well-defined. To estimate $\|(J_f(x^{(0)}))^{-1}\|$, we first find

$$\|(J_f(x^*)^{-1}[J_f(x^{(0)}) - J_f(x^*)])\| \leq CLr \leq \frac{1}{2}.$$

In particular,

$$\|(J_f(x^{(0)}))^{-1}\| \leq \frac{\|(J_f(x^*))^{-1}\|}{1 - \|(J_f(x^*)^{-1}[J_f(x^{(0)}) - J_f(x^*)])\|} \leq 2C.$$

Using this, we deduce

$$\begin{aligned} x^{(1)} - x^* &= x^{(0)} - x^* - (J_f(x^{(0)}))^{-1} f(x^{(0)}) \\ &= x^{(0)} - x^* - (J_f(x^{(0)}))^{-1} [f(x^{(0)}) - f(x^*)] \\ &= (J_f(x^{(0)}))^{-1} [(f(x^*) - f(x^{(0)})) - J_f(x^{(0)})(x^* - x^{(0)})]. \end{aligned}$$

Next, consider the Taylor expansion of f around $x^{(0)}$ at x^* :

$$f(x^*) - f(x^{(0)}) = J_f(x^{(0)})(x^* - x^{(0)}) + R,$$

where R encodes the second-derivative information of f . The important part is that f being L -smooth means R is bounded in terms of L and the squared distance between x^* and $x^{(0)}$, i.e.,

$$\|R\| \leq \frac{L}{2} \|x^* - x^{(0)}\|^2.$$

Taking these facts together yields

$$\|x^{(1)} - x^*\| \leq \|(J_f(x^{(0)}))^{-1}\| \|R\| \leq CL \|x^* - x^{(0)}\|^2.$$

In particular, $\|x^* - x^{(0)}\| \leq r \leq \frac{1}{2CL}$, and so $\|x^{(1)} - x^*\| \leq \frac{1}{2} \|x^* - x^{(0)}\| < r$.

So, $x^{(1)} \in B_r(x^*)$, and by following a simple induction argument, we see that $x^{(k)} \in B_r(x^*)$ for all k , with

$$\|x^* - x^{(k)}\| \leq CL \|x^* - x^{(k-1)}\|^2 \leq \frac{r}{2^k}.$$

From both inequalities above, we see that $x^{(k)} \rightarrow x^*$, and that the convergence is quadratic.

(c)

If we are minimizing, say, a convex, C^3 function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, then Newton's method for minimizing f amounts to applying Newton's method for solving the nonlinear system $\nabla f(x) = 0$, i.e., given $x^{(0)}$, the iterates are

$$x^{(k+1)} = x^{(k)} - H_f(x^{(k)})^{-1} \nabla f(x^{(k)}).$$

Assuming the conditions outlined in part (a) hold for ∇f , our proof from before will establish that the Newton iterates converge quadratically to a root of ∇f , which is therefore a minimum of f by convexity.

With the addition of constraints of the form $h(x) = 0$, Newton's method for minimization will converge provided h is sufficiently smooth. This is because we now consider the unconstrained minimization of the penalized objective $f(x) + \alpha \|h(x)\|^2$. If h has the same regularity as f , then Newton's method will converge for any α .

Problem 3

(a)

The maximum degree of exactness is at least 1, which corresponds to using the trapezoidal rule, i.e., $a = b = \frac{h}{2}$. We claim that if this quadrature rule has degree of exactness at least 1, it must be the trapezoidal rule. To see this, test with $f(x) = 1$ and $f(x) = x$:

$$\begin{aligned} \int_0^h 1 \, dx &= h = a + b, \\ \int_0^h x \, dx &= \frac{h^2}{2} = bh. \end{aligned}$$

The second equation tells us $b = \frac{h}{2}$, and the first equation tells us $a = \frac{h}{2}$ as well. Hence, the maximum degree of exactness is the degree of exactness of the trapezoidal rule, namely, 1, since the trapezoidal rule is not exact for $f(x) = x^2$:

$$\int_0^h x^2 \, dx = \frac{h^3}{3} \neq \frac{h}{2}(h^2)$$

(b)

Start by maximizing the degree of exactness. We have three degrees of freedom, so we might expect a degree of exactness of 2. By the linearity of integration,

we only need to test with 1, x , and x^2 to get a system of equations

$$\begin{aligned} h &= a + b, \\ \frac{h^2}{2} &= bc, \\ \frac{h^3}{3} &= bc^2. \end{aligned}$$

Solving gives $a = \frac{h}{4}$, $b = \frac{3h}{4}$, and $c = \frac{2h}{3}$. The degree of exactness is exactly 2, since if we test with x^3 , we get

$$\int_0^h x^3 dx = \frac{h^4}{4} \neq \frac{3h}{4} \cdot \frac{8h^3}{27} = \frac{2h^4}{9}.$$

To find the order of accuracy with this method, assume $f \in C^3[0, h]$ and Taylor expand f around 0:

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2}x^2 + \frac{f'''(\xi_x)}{6}x^3.$$

Integrating gives

$$\int_0^h f(x) dx = hf(0) + \frac{h^2}{2}f'(0) + \frac{h^3}{6}f''(0) + \int_0^h \frac{f'''(\xi_x)}{6}x^3 dx.$$

On the other hand,

$$bf(c) = bf(0) + bcf'(0) + \frac{bc^2}{2}f''(0) + \frac{bc^3}{6}f'''(\xi_c),$$

and so

$$af(0) + bf(c) = hf(0) + \frac{h^2}{2}f'(0) + \frac{h^3}{6}f''(0) + \frac{h^4}{27}f'''(\xi_c).$$

Taking these facts together yields

$$\int_0^h f(x) dx - (af(0) + bf(c)) = \int_0^h \frac{f'''(\xi_x)}{6}x^3 dx + \frac{h^4}{27}f'''(\xi_c).$$

Since $f \in C^3[0, h]$, the third derivative term is bounded, so the integral on the right hand side is $O(h^4)$. Hence, the order of accuracy is 4.

(c)

Test with the following step function:

$$f(x) = \begin{cases} 0 & x \in [0, h/2], \\ 1 & x \in [h/2, h] \end{cases}.$$

In this case, $\int_0^h f(x) dx = \frac{h}{2}$, while the quadrature rule gives us $\frac{3h}{4}$. This tells us the highest order of accuracy is 1, i.e., this quadrature rule cannot have order of accuracy greater than 1 over this class of discontinuous functions.

August 2019

Problem 1

(a)

From August 2023, problem 1, we know what Cholesky factorization looks like. For solving a linear system $Ax = b$, Cholesky factorization plays a similar role to LU factorization. In particular, if $A = H^T H$ is the Cholesky factorization of A , one solves $Ax = b$ by solving in order the triangular systems $H^T y = b$ and $Hx = y$.

Again, from August 2023, forming the Cholesky factor takes $\frac{1}{6}(n^3 + 3n^2 - 4n)$ multiplications. The number of additions is on the same order as the number of multiplications (in fact, they differ by just $n - 1$), so the computational cost of Cholesky factorization is on the order of $\frac{1}{3}n^3$. In contrast, LU factorization has a computational cost on the order of $\frac{2}{3}n^3$ (this is because Gaussian elimination requires that many operations).

(b)

See problem 2(a) of January 2015 for a full derivation of the normal equations.

If A is full rank, then $A^T A$ is symmetric and positive definite, meaning it has a Cholesky decomposition $H^T H$. This means the normal equations can be reformulated as

$$H^T Hx = A^T b.$$

This can be solved by solving (in order) the two systems

$$\begin{aligned} H^T y &= A^T b, \\ Hx &= y. \end{aligned}$$

Since H is upper triangular, both systems can be easily solved via back-substitution.

Due to the requirement of A being full rank, this method does not always work.

As an example, if $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$, then $A^T A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, which is not positive definite and therefore has no Cholesky decomposition.

(c)

The technique is explained in January 2024, problem 1(a). The pseudoinverse is advantageous when there are infinitely many solutions to the normal equations;

this can occur for underdetermined systems or overdetermined systems with A not having full rank. In both of these cases, the pseudoinverse will select the least-squares solution with minimum norm.

Problem 2

(a)

The definition is January 2024, problem 2(a). Convergence is proven similarly to January 2020, 2b, but much simpler in the scalar case. Assume f is C^2 with bounded derivatives on some neighborhood x^* and that $f'(x^*) \neq 0$. First, we find

$$\begin{aligned} x^{(k+1)} - x^* &= x^{(k)} - x^* - \frac{f(x^{(k)})}{f'(x^{(k)})} \\ &= \frac{1}{f'(x^{(k)})} \left[f'(x^{(k)})(x^{(k)} - x^*) - f(x^{(k)}) \right] \end{aligned}$$

Taylor expanding f about $x^{(k)}$ yields

$$0 = f(x^*) = f(x^{(k)}) + f'(x^{(k)})(x^* - x^{(k)}) + f''(\xi^{(k)})(x^* - x^{(k)})^2$$

for some $\xi^{(k)}$ between x^* and $x^{(k)}$, and so

$$f'(x^{(k)})(x^{(k)} - x^*) - f(x^{(k)}) = f''(\xi^{(k)})(x^{(k)} - x^*)^2.$$

Taking these facts together gives us

$$\left| x^{(k+1)} - x^* \right| = \left| \frac{f''(\xi^{(k)})}{f'(x^{(k)})} \right| \left| x^{(k)} - x^* \right|^2.$$

Since $f'(x^*) \neq 0$, there is some neighborhood $B_r(x^*)$ around x^* such that the quotient $\left| \frac{f''}{f'} \right|$ is bounded by some constant $C \geq 0$. Finally, if $s = \min(r, (2C)^{-1})$ and $x^{(0)} \in B_s(x^*)$, it follows from induction that $x^{(k)} \in B_s(x^*)$ for all k and that $x^{(k)} \rightarrow x^*$. Moreover, the estimate above tells us that the convergence is quadratic.

(b)

January 2024, problem 2(c).

(c)

There are multiple options here, but the most well-known are the bisection and secant methods.

Recall that the bisection method requires that f be continuous and there exist

two points $a < b$ such that $f(a)f(b) < 0$. In this case, let $a^{(0)} = a$, $b^{(0)} = b$, and $x^{(0)} = \frac{a^{(0)} + b^{(0)}}{2}$. Then, for $k \geq 0$, put

$$\begin{cases} a^{(k+1)} = a^{(k)}, b^{(k+1)} = x^{(k)} & \text{if } f(a^{(k)})f(x^{(k)}) < 0, \\ a^{(k+1)} = x^{(k)}, b^{(k+1)} = b^{(k)} & \text{otherwise,} \end{cases}$$

$$x^{(k+1)} = \frac{a^{(k+1)} + b^{(k+1)}}{2}.$$

The secant method is akin to Newton's method, but uses a secant line approximation of the derivative. Precisely, given two initial guesses $x^{(0)}$ and $x^{(1)}$, we have for $k \geq 1$

$$x^{(k+1)} = x^{(k)} - \frac{x^{(k)} - x^{(k-1)}}{f(x^{(k)}) - f(x^{(k-1)})} f(x^{(k)})$$

Problem 3

(a)

Something tells me that the quadrature rule is supposed to be $af(0) + bf(h)$, since this does not make much sense if $h < 1$. Nonetheless, I will give the benefit of the doubt and write the solution with the form given.

To maximize the degree of exactness, test with 1 and x to get the system

$$\begin{aligned} \int_{-h}^h 1 \, dx &= 2h = a + b, \\ \int_{-h}^h x \, dx &= 0 = b. \end{aligned}$$

Thus, $a = 2h$ and $b = 0$, so the rule reduces to the midpoint rule. In August 2020, problem 3(a), we found that the order of accuracy for the midpoint rule is 3.

(b)

August 2020, problem 3(b)

(c)

Identical to August 2020, problem 3(c), but $\mathcal{A}(h)$ now denotes the composite trapezoidal rule with subinterval length h rather than the composite midpoint rule.

January 2017

Problem 1

August 2023, problem 1

Problem 2

(a)

August 2023

(b)

August 2023

(c)

If the minimization is constrained such that $x \in \Omega \subseteq \mathbb{R}^d$, describe a penalty method for the minimization of the constrained problem.

Assuming Ω is the zero set of some function h , i.e., $\Omega = \{x \in \mathbb{R}^d : h(x) = 0\}$, then the penalty method solves

$$\text{minimize} \left(f(x) + \frac{\alpha_k}{2} \|h(x)\|_2^2 \right)$$

for all k , where (α_k) is a sequence of positive scalars that increase to $+\infty$. The minimizers x_k^* converge to the constrained minimizer provided Ω is closed and f and h are continuous on Ω .

Problem 3

(a)

August 2023, problem 3(a).

(b)

August 2020, problem 3(b)

(c)

First, observe that $f''(x) = e^{2x}(3\sin x + 4\cos x)$, so we can coarsely bound the second derivative with $\|f''\|_{L^\infty(-1,1)} = 7e^2$. Recall that the error for the midpoint rule over an interval of length h is

$$E_0(f) = \frac{h^3}{3} f''(\xi)$$

for some ξ inside this interval. So, the error over all $2n$ subintervals is

$$E_{0,2n}(f) = \sum_{i=1}^{2n} \frac{1}{3n^3} f''(\xi_i),$$

and thus

$$|E_{0,2n}(f)| \leq \|f''\|_{\infty} \sum_{i=1}^{2n} \frac{1}{3n^3} \leq 7e^2 \frac{2n}{3n^3} = \frac{14e^2}{3} \cdot \frac{1}{n^2}.$$

To attain an accuracy of 10^{-4} , we require

$$n \geq \sqrt{\frac{14e^2}{3 \cdot 10^{-4}}}.$$

January 2016

Problem 1

(a)

Suppose $Ax = 0$ with $x \neq 0$ and let x_i be the component of x with the largest absolute value (of course, $x_i \neq 0$). Then,

$$\begin{aligned} Ax = 0 &\implies \sum_{j=1}^n a_{ij}x_j = a_{ii}x_i + \sum_{\substack{j=1 \\ j \neq i}}^n a_{ij}x_j = 0 \\ &\implies a_{ii} = - \sum_{\substack{j=1 \\ j \neq i}}^n \frac{x_j}{x_i} a_{ij} \\ &\implies |a_{ii}| \leq \sum_{\substack{j=1 \\ j \neq i}}^n \frac{|x_j|}{|x_i|} |a_{ij}| \leq \sum_{\substack{j=1 \\ j \neq i}}^n |a_{ij}|, \end{aligned}$$

a contradiction. Thus, A is nonsingular. If instead A was strictly diagonally dominant by columns, then this argument can be repeated with A^T , noting that A is nonsingular if and only if A^T is.

(b)

Defined in January 2020, problem 1(a). If A is strictly diagonally dominant (by rows), then

$$\begin{aligned}
\|B_J\|_\infty &= \|I - D^{-1}A\|_\infty \\
&= \max_{i=1,\dots,n} \left(\sum_{\substack{j=1 \\ j \neq i}}^n \left| \frac{a_{ij}}{a_{ii}} \right| \right) \\
&= \max_{i=1,\dots,n} \left(\frac{1}{|a_{ii}|} \sum_{\substack{j=1 \\ j \neq i}}^n |a_{ij}| \right) \\
&< 1,
\end{aligned}$$

where D is the diagonal of A .

Next, suppose A is strictly diagonally dominant by columns. Defining $\tilde{B}_J = (DB_J D^{-1})^T$, we have

$$\begin{aligned}
\tilde{B}_J &= D^{-1} B_J^T D \\
&= D^{-1} (I - D^{-1} A)^T D \\
&= D^{-1} (I - A^T D^{-1}) D \\
&= (D^{-1} - D^{-1} A^T D^{-1}) D \\
&= I - D^{-1} A^T.
\end{aligned}$$

Notice that this is the iteration matrix of the Jacobi method for solving a system of the form $A^T x = b$. Since A^T is strictly diagonally dominant by *rows*, we must have $\rho(\tilde{B}_J) < 1$ by the above case. But since similarity transforms and transposition do not affect the spectrum of a square matrix, we have $\rho(B_J) < 1$ as well.

(c)

For an overdetermined system, the normal equations read off as

$$A^T A x = A^T b,$$

while the pseudoinverse is

$$x^* = A^\dagger b = V \Sigma^\dagger U^T b,$$

where $A = U \Sigma V^T$ is an SVD of A , and

$$\Sigma^\dagger = \text{diag}(\sigma_1^{-1}, \dots, \sigma_r^{-1}, 0, \dots, 0)$$

for $\sigma_1, \dots, \sigma_r$ the nonzero singular values of A (in descending order). If A has full rank, then $A^\dagger = (A^T A)^{-1} A^T$, which implies that the normal equations have a unique solution that agrees with $A^\dagger b$. If A does not have full rank, then $A(A^\dagger b + z) = AA^\dagger b$ for any $z \in \text{Null}(A)$, meaning $A^\dagger b + z$ is also a least-squares solution. In other words, the pseudoinverse is always a least-squares solution, but not all least-squares solutions are the pseudoinverse.

Problem 2

(a)

August 2023, problem 2(a).

(b)

August 2023, problem 2(b) applies here as well, even without the additional third derivative condition.

(c)

The penalty method formulation is January 2017, problem 2(c).

I'm not sure about this problem. Surely, convergence should depend on what Ω is, and I'm not sure how this relates at all to f' being positive.

Problem 3

(a)

Let $\mathcal{I} = [x_0, x_{N-1}]$. The interpolation error on $\mathcal{I}$ is bounded by the maximum of the interpolation errors on all subintervals. The following estimate will hold uniformly over all subintervals, so we can assume without loss of generality that we have $n = 1$ subinterval. From part (b), we have the error formula

$$f(x) - \Pi_p f(x) = \frac{f^{(p+1)}(\xi)}{(p+1)!} \prod_{i=0}^p (x - x_i)$$

for some $\xi \in (x_0, x_{N-1})$. Now, f being C^∞ means $f^{(p+1)}$ is uniformly bounded on $\mathcal{I}$, and $|x - x_i| \leq x_{N-1} - x_0 \leq ph$ for $x \in \mathcal{I}$ and $i = 0, \dots, N-1$, so

$$|f(x) - \Pi_p f(x)| \leq \frac{\|f^{(p+1)}\|_{L^\infty(\mathcal{I})}}{(p+1)!} (ph)^{p+1}.$$

The right hand side tends to 0 as $h \rightarrow 0$.

(b)

We will derive the standard formula for the interpolation error from Lagrange interpolation, and once again, we only do this for one subinterval. Fix $x \in \mathcal{I}$. If x is any of the interpolation nodes, then the formula trivially holds since both sides must be zero. So, assume x is not any of the interpolation nodes. Then, define $E_p(t) := f(t) - \Pi_p f(t)$ and $\omega_p(t) = \Pi_{i=0}^p (t - x_i)$ on $\mathcal{I}$, and let

$$G(t) = E_p(t) - \frac{\omega_p(t)E_p(x)}{\omega_p(x)}.$$

Note that $G \in C^\infty$ since f and ω_p are both C^∞ . Moreover, G has at least $p+2$ roots, namely, $x_0, \dots, x_p$, and x . It follows that $G^{(p+1)}$ has at least one root, which we will denote by ξ . Now,

$$G^{(p+1)}(t) = f^{(p+1)}(t) - (p+1)! \frac{E_p(x)}{\omega_p(x)}$$

since $\Pi_p f(t)$ has degree p and the leading term of $\omega_p(t)$ is t^{p+1} . From here, plug in ξ and rearrange to obtain

$$E_p(x) = \frac{f^{(p+1)}(\xi)}{(p+1)!} \omega_p(x).$$

(c)

Interpolation error in H^1 is not discussed in this book; this is likely an outdated problem that will not show up in the future.

August 2015

Problem 1

(a)

Assuming that the method above is *consistent* (meaning $f = x - Bx$), then it converges for any $x^{(0)}$ if and only if $\rho(B) < 1$.

To see this, let $e^{(k)} = x^{(k)} - x$, so that

$$Be^{(k)} = Bx^{(k)} - Bx = x^{(k+1)} - f - (x - f) = e^{(k+1)}.$$

Thus, $e^{(k)} = B^k e^{(0)}$ for all k . But now, $\rho(B) < 1 \iff B^k e^{(0)} \rightarrow 0$ for all $e^{(0)} \iff x^{(k)} \rightarrow x$ for all choices of $x^{(0)}$.

(b)

January 2016, problem 1(b)

(c)

The iteration matrix is $B_J = I - D^{-1}A$. If A is upper triangular, then B_J is upper triangular with zeros along the diagonal. Such matrices are nilpotent, i.e., $B_J^m = 0$ for some integer m . So, $e^{(k)} = 0$ for $k \geq m$, meaning the Jacobi method converges within m steps.

Problem 2

(a)

August 2019, problem 2(a), but now for finding the root of ∇f .

(b)

We see that $\Omega = \{x \in \mathbb{R}^2 : x_1^2 + x_2^2 \leq 1\}$, so we can write the Lagrangian as

$$\mathcal{L}(x_1, x_2, \lambda) = f(x) + \lambda(x_1^2 + x_2^2 - 1).$$

If $x^* = (x_1^*, x_2^*)$ is a constrained minimum of f , then there exists $\lambda^* \in \mathbb{R}$ such that the following Kuhn-Tucker conditions are satisfied:

1. $(x_1^*)^2 + (x_2^*)^2 \leq 1$
2. $\lambda^* \geq 0$
3. $\nabla_x \mathcal{L}(x_1^*, x_2^*, \lambda^*) = \nabla f(x^*) + 2\lambda^* x^* = 0$
4. $\lambda^* ((x_1^*)^2 + (x_2^*)^2 - 1) = 0$

(c)

August 2023, problem 2(c), but now use the penalty function $h(x_1, x_2) = \max(x_1^2 + x_2^2 - 1, 0)$.

Problem 3

(a)

If p_1 and p_2 were two polynomials of degree $n - 1$ that agreed on n points, then $p_1 - p_2$ is a polynomial of degree at most $n - 1$ that has n roots. By the fundamental theorem of algebra, this is only possible if $p_1 - p_2 = 0$.

(b)

Consider, for example, the n monomials $\{x, x^3, \dots, x^{2n-1}\}$. No linear combination of these could interpolate, say, the points $(0, 1), (1, 1), \dots, (n, 1)$, since any linear combination of these monomials evaluated at 0 is 0.

To see that uniqueness is not guaranteed, consider the case $n = 2$, i.e., our

monomials are x and x^3 . Then, there are several linear combinations of x and x^3 that interpolate $(0, 0)$ and $(1, 1)$, for example, $\frac{1}{2}x + \frac{1}{2}x^3$ and x .

(c)

To my knowledge, Fourier interpolation is not a Numerical I topic.

January 2015

Problem 1

(a)

We find that the eigenvalues of $A^T A$ are 200 and 50, so the singular values are $10\sqrt{2}$ and $5\sqrt{2}$. So, we let

$$\Sigma = \begin{pmatrix} 10\sqrt{2} & 0 \\ 0 & 5\sqrt{2} \end{pmatrix}.$$

Next, we find that $(3, -4)^T$ and $(4, 3)^T$ are eigenvectors of $A^T A$ corresponding to 200 and 50 respectively. These vectors are already orthogonal, so we are motivated to define

$$V = \begin{pmatrix} 3/5 & 4/5 \\ -4/5 & 3/5 \end{pmatrix},$$

where we divide by each entry of V by 5 to ensure it is unitary. To find U , we simply multiply

$$U = AV\Sigma^{-1} = \begin{pmatrix} -1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}.$$

This gives us our SVD

$$A = U\Sigma V^T = \begin{pmatrix} -1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 10\sqrt{2} & 0 \\ 0 & 5\sqrt{2} \end{pmatrix} \begin{pmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{pmatrix}.$$

To reduce the number of minus signs, we can multiply the first column of U and the first row of V^T by -1 ; doing so does not affect the final product, so

$$A = U\Sigma V^T = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 10\sqrt{2} & 0 \\ 0 & 5\sqrt{2} \end{pmatrix} \begin{pmatrix} -3/5 & 4/5 \\ 4/5 & 3/5 \end{pmatrix}.$$

Notice that U and V are unitary.

(b)

Let u_1, u_2 be the columns of U , v_1, v_2 be the columns of V , and σ_1, σ_2 the diagonal entries of Σ from part (a). Since $A^T U = V \Sigma$ and $AV = U \Sigma$, we have $A^T u_i = \sigma_i v_i$ and $Av_i = \sigma_i u_i$ for $i = 1, 2$. It follows that

$$B \begin{pmatrix} v_i \\ u_i \end{pmatrix} = \begin{pmatrix} A^T u_i \\ Av_i \end{pmatrix} = \begin{pmatrix} \sigma_i v_i \\ \sigma_i u_i \end{pmatrix} = \sigma_i \begin{pmatrix} v_i \\ u_i \end{pmatrix}.$$

Likewise,

$$B \begin{pmatrix} v_i \\ -u_i \end{pmatrix} = \sigma_i \begin{pmatrix} -v_i \\ u_i \end{pmatrix} = -\sigma_i \begin{pmatrix} v_i \\ -u_i \end{pmatrix}.$$

Thus, the eigenvalues of B are $\pm\sigma_1, \pm\sigma_2$, and the corresponding unit eigenvectors are $\frac{1}{\sqrt{2}} \begin{pmatrix} v_1 \\ \pm u_1 \end{pmatrix}$ and $\frac{1}{\sqrt{2}} \begin{pmatrix} v_2 \\ \pm u_2 \end{pmatrix}$. This gives us an eigenvalue decomposition

$$\begin{aligned} B &= \frac{1}{2} \begin{pmatrix} v_1 & v_2 & v_1 & v_2 \\ u_1 & u_2 & -u_1 & -u_2 \end{pmatrix} \begin{pmatrix} \sigma_1 & 0 & 0 & 0 \\ 0 & \sigma_2 & 0 & 0 \\ 0 & 0 & -\sigma_1 & 0 \\ 0 & 0 & 0 & -\sigma_2 \end{pmatrix} \begin{pmatrix} v_1 & v_2 & v_1 & v_2 \\ u_1 & u_2 & -u_1 & -u_2 \end{pmatrix}^T \\ &= \frac{1}{2} \begin{pmatrix} V & V \\ U & -U \end{pmatrix} \begin{pmatrix} \Sigma & 0 \\ 0 & -\Sigma \end{pmatrix} \begin{pmatrix} V & V \\ U & -U \end{pmatrix}^{-1} \end{aligned}$$

Problem 2

(a)

Write

$$\begin{aligned} \|Ax - b\|_2^2 &= (Ax - b)^T (Ax - b) = \|Ax\|_2^2 - 2\langle Ax, b \rangle + \|b\|_2^2 \\ &= x^T A^T A x - 2x^T A^T b + b^T b. \end{aligned}$$

Taking the gradient in x yields

$$\nabla_x \|Ax - b\|_2^2 = 2A^T A x - 2A^T b.$$

The above gradient is zero if and only if

$$A^T A x = A^T b,$$

that is, the normal equations. Note that $\|Ax - b\|_2^2$ is convex as a function of x , so a solution to the normal equations minimizes $\|Ax - b\|_2$.

(b)

The solution with QR decomposition is August 2020, problem 1(c). The solution with SVD is January 2024, problem 1(a).

(c)

If A is not full rank, the least-squares solution need not be unique. Consider $A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$ and $b = (1, 0, 0)$. Then, $(1, 0)$ and $(0, 1)$ are both least-squares solutions (in fact, they are exact solutions to $Ax = b$).

In general, the least-squares solutions are of the form

$$A^\dagger b + z = V\Sigma^\dagger U^T b + z$$

for any $z \in \text{Null}(A)$.

(d)

Given a tolerance $\varepsilon > 0$, let r be the first index for which $\sigma_{r+1} < \varepsilon$. Since all singular values after σ_{r+1} are smaller than this tolerance, we can truncate A to its first r singular values. That is, we now use the matrix $\tilde{\Sigma} \in \mathbb{R}^{m \times n}$ in place of Σ , where $\tilde{\Sigma} = \text{diag}(\sigma_1, \dots, \sigma_r, 0, \dots, 0)$. Then, we approximate the Moore-Penrose pseudoinverse by $(\tilde{A})^\dagger := V(\tilde{\Sigma})^\dagger U^T$, where $(\tilde{\Sigma})^\dagger = \text{diag}(\sigma_1^{-1}, \dots, \sigma_r^{-1}, 0, \dots, 0)$.

This will give an approximate solution to the least-squares problem while avoiding numerical stability concerns that come from multiplying by the inverses of many small singular values.

Problem 3

The best method is still Newton's method, which can be derived by looking at a Taylor expansion of ∇G around some iterate $x^{(k)}$, namely,

$$\nabla G(x^{(k+1)}) = \nabla G(x^{(k)}) + H_G(x^{(k)})(x^{(k+1)} - x^{(k)}) + O\left(\|x^{(k+1)} - x^{(k)}\|^2\right),$$

where $O(\|x^{(k+1)} - x^{(k)}\|^2)$ is some second-order term. We discard the second order term and approximate $\nabla G(x^{(k+1)}) \approx 0$ (since we want $x^{(k+1)}$ to be close to a root of ∇G) and obtain

$$0 \approx \nabla G(x^{(k)}) + H_G(x^{(k)})(x^{(k+1)} - x^{(k)}).$$

If we treat this approximation as an equality to define our iterates, then by rearranging, we have the familiar Newton's method iterates

$$x^{(k+1)} = x^{(k)} - H_G(x^{(k)})^{-1} \nabla G(x^{(k)}).$$

If G is quadratic, then Newton's method converges in 1 step for any initial guess $x^{(0)}$. This is because G being quadratic means $\nabla G(x)$ is affine, i.e.,

$\nabla G(x) = Ax + b$ for some matrix A and some vector b , and consequently, the Hessian is a constant $H_G(x) \equiv A$. With this, the first iterate looks like

$$x^{(1)} = x^{(0)} - H_G^{-1} \nabla G(x^{(0)}) = x^{(0)} - A^{-1}[Ax^{(0)} + b] = -A^{-1}b,$$

and so

$$\nabla G(x^{(1)}) = A[-A^{-1}b] + b = 0.$$

Since G is strictly convex, $\nabla G(x^{(1)})$ being zero means $x^{(1)}$ is the global minimum of G .